

7.2

prostor V má standardní bázi $S = (1, x, x^2)$

$$\phi(p) = 2p' + x^2 p''$$

$$\phi(1) = 2(0) + \cancel{0} x^2(0) = 0$$

$$\phi(x) = 2 + 0 = 2$$

$$\phi(x^2) = 4x + 2x^2 = 2x^2 + 4x$$

$$[\phi]_S = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = \lambda^2(2-\lambda)$$

$\begin{matrix} 0 & 2 \\ \uparrow & \uparrow \\ \lambda^2 & (2-\lambda) \end{matrix}$

pro $\lambda = 0$ matice $A - 0I$ má hodnost 2 tedy $\dim(\ker \phi) = 3 - 2 = 1$

pro $\lambda = 0$ máme tedy 1 lineární vektor

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$B = (v_1, v_2, v_3)$$

$$1. \phi(v_2) = v_1 \quad \text{a} \quad \phi(v_1) = 0$$

$$2. \phi(v_3) = 2v_3$$

oblatni vektor pro $\lambda = 2$

$$(A - 2I)V = 0$$

$$\begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} z = 1 \Rightarrow y = 2 \quad x = 2 \\ v_3 = 2x^2 + 2y + 2 = x^2 + y + 1 \end{matrix}$$

$$v_1 \in \ker(\phi): Av = 0$$

$$v_1 = 1 \quad \phi(v_2) = 1$$

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$2b = 1 \Rightarrow b = \frac{1}{2}$$

$$4c = 0 \Rightarrow c = 0$$

$$\text{tedy } v_2 = \frac{1}{2}x$$

$$a = 0$$

$$\text{máme tedy bázi } (1, \frac{1}{2}x, x^2 + x + 1)$$

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\text{báze } B = (1, \frac{1}{2}x, x^2 + x + 1)$$